

AI Governance Risk Assessment

AI-Assisted Resume Screening Systems in Canadian Hiring

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This document is an independent governance case study prepared for illustrative and professional portfolio purposes.

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1. Executive Summary

This document presents an AI governance risk assessment of AI-assisted resume screening systems commonly used in Canadian hiring contexts, with representative examples including LinkedIn Recruiter and Workday Recruiting. These systems support recruiters by analyzing resumes and job descriptions to prioritize and rank candidates, thereby influencing early-stage hiring decisions.

The use of AI-assisted resume screening has increased in response to high application volumes, efficiency pressures, and the digitization of recruitment practices. Because these systems shape access to employment opportunities, they raise important governance considerations related to fairness, transparency, accountability, and stakeholder trust. In the Canadian context, these considerations are particularly salient given human rights protections, diversity and inclusion commitments, and growing expectations around responsible AI use in organizational decision-making.

This assessment identifies several key governance risks associated with AI-assisted resume screening, including the potential reinforcement of historical hiring biases, reduced visibility for candidates with non-traditional career paths or from underrepresented groups, over-reliance on automated rankings by recruiters, and limited transparency regarding how candidate prioritization occurs. While job applicants are the primary stakeholders affected, these risks also have implications for recruiters, organizations, and broader labor-market trust.

To address these risks, the assessment emphasizes governance controls that preserve human oversight, clearly define system scope and limitations, and support ongoing monitoring and review. Recommended practices include human-in-the-loop decision processes, documentation of intended use and constraints, periodic bias, and performance reviews, and clearly assigned accountability within recruitment workflows.

This assessment is informed by the structure and principles of the NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0), particularly the Govern, Map, Measure, and Manage functions. The objective is not to evaluate technical performance or vendor compliance, but to demonstrate how structured, stakeholder-aware governance practices can support the responsible deployment of AI systems in hiring contexts.

2. Purpose, Scope, and Methodology

2.1 Purpose of the Assessment

The purpose of this assessment is to examine governance considerations associated with the use of AI-assisted resume screening systems in Canadian hiring contexts. These systems increasingly influence how candidates are prioritized and shortlisted during preliminary stages of recruitment, shaping access to employment opportunities and organizational decision-making.

AI-assisted resume screening represents a relevant governance case due to its widespread adoption, its direct impact on individuals and labor-market outcomes, and its operation within a regulated social and legal environment. In Canada, hiring practices are subject to human rights protections, employment equity considerations, and growing expectations around transparency and accountability in automated decision-support systems. This assessment seeks to identify and articulate governance risks associated with these systems and to highlight practices that can support their responsible and trustworthy use.

2.2 Scope

This assessment focuses on the governance, risk, and oversight dimensions of AI-assisted resume screening systems commonly embedded in enterprise recruitment platforms. The review considers system purpose, deployment context, stakeholder impacts, and organizational controls relevant to the use of AI in candidate prioritization.

The assessment is in scope for:

- AI-assisted resume analysis, matching, and ranking used to support recruiter decision-making.
- Governance risks related to fairness, transparency, accountability, and stakeholder trust.
- Organizational practices that influence responsible deployment and use

The assessment is out of scope for:

- Evaluation of proprietary algorithms, model architecture, or source code
- Validation of technical performance metrics or predictive accuracy
- Vendor-specific compliance audits or certification assessments

This scoping reflects the intent to assess governance practices rather than technical implementation details.

2.3 Methodology

The assessment adopts a qualitative, stakeholder-informed governance approach aligned with the structure and principles of the NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0). The methodology emphasizes contextual understanding, risk identification, and practical governance considerations rather than technical evaluation.

Key elements of the methodology include:

- High-level system analysis based on publicly documented functionality and common deployment practices.
- Identification of relevant stakeholders and assessment of their relative salience and exposure to risk
- Qualitative identification of governance risks across fairness, transparency, accountability, and trust dimensions
- Consideration of governance controls and mitigation practices consistent with responsible AI principles

The assessment is informed by the NIST AI RMF functions (Govern, Map, Measure, and Manage) to structure analysis and ensure consistency with established AI risk management practices.

3. System Overview

3.1 System Description

The system assessed in this document is an AI-assisted resume screening capability commonly embedded within enterprise recruitment platforms used in Canadian hiring contexts, such as LinkedIn Recruiter and Workday Recruiting. The system is designed to support early-stage recruitment by analyzing candidate resumes in relation to job requirements.

At a high level, the system ingests structured and unstructured data, including resumes, candidate profiles, and job descriptions. Using machine learning and natural language processing techniques, it identifies relevant skills, experience, and keywords, and produces outputs such as relevance scores, ranked candidate lists, or recommendations intended to assist recruiters in prioritizing applications for further review. Figure 1

provides a high-level representation of the system and how AI-assisted resume screening integrates into typical recruitment workflows. The system does not make final hiring decisions. Instead, it provides decision-support outputs that inform human review and subsequent recruitment actions.

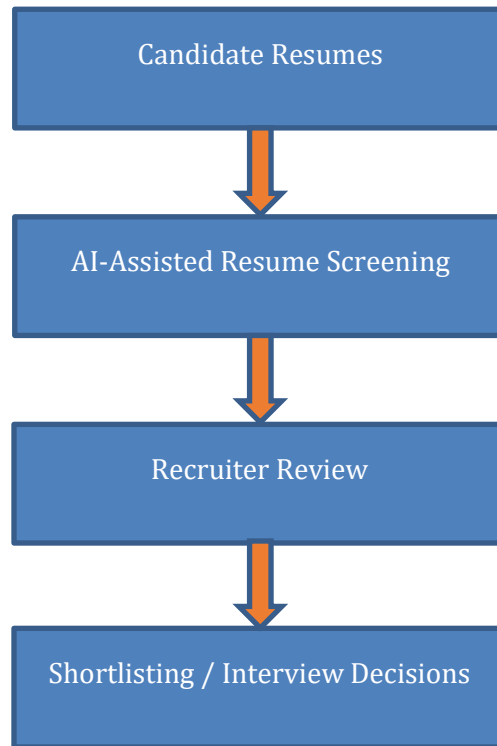


Figure 1. High-level workflow of AI-assisted resume screening within the recruitment processes.

3.2 Intended Use

The intended use of the AI-assisted resume screening system is to support recruiters and hiring teams by improving efficiency in managing high volumes of applications and by assisting with the identification of potentially suitable candidates. The system is designed to function as a screening and prioritization aid within broader recruitment workflows.

Human judgment remains central to the hiring process. Recruiters are expected to review AI-generated outputs, apply contextual understanding, and exercise discretion when shortlisting candidates. The system is not intended to replace human decision-making, nor is it intended to operate autonomously or without oversight. Use of the system outside its intended scope (for example, as a sole decision-maker or as an automated exclusion mechanism) would raise significant governance concerns.

3.3 Deployment Context

In Canada, AI-assisted resume screening systems are deployed within organizational environments that are shaped by legal, social, and ethical considerations related to employment practices. Hiring decisions are subject to federal and provincial human rights legislation, employment equity principles, and organizational commitments to diversity, equity, and inclusion.

Canadian organizations increasingly operate in competitive labor markets characterized by digital recruitment channels and large applicant pools. Within this context, AI-assisted screening tools are often adopted to improve efficiency and consistency. However, their deployment also occurs within a broader societal context that places emphasis on fairness, transparency, and accountability in automated or algorithm-supported decision processes. These contextual factors make AI-assisted resume screening a particularly relevant use case for governance analysis, as system design and use may influence not only organizational outcomes but also public trust in hiring practices and labor-market institutions.

4. Stakeholder Identification and Salience

4.1 Key Stakeholders

AI-assisted resume screening systems affect a range of stakeholders across the recruitment lifecycle, each experiencing various levels of influence, exposure, and risk. Figure 2 illustrates key stakeholder groups and their relative proximity to system impacts.

Primary stakeholders include job applicants and recruiters. Job applicants are directly affected by how their resumes are analyzed, ranked, and surfaced for human review, as these processes shape access to employment opportunities. Recruiters and hiring managers are primary users of the system and rely on its outputs to support early-stage decision-making under time and volume constraints.

Secondary stakeholders include employing organizations, human resources leadership, and diversity, equity, and inclusion (DEI) teams. Organizations are affected by the quality, fairness, and consistency of hiring outcomes, as well as by potential reputational and compliance risks. HR leadership and DEI teams have a vested interest in ensuring that recruitment practices align with organizational values, employment equity objectives, and legal obligations.

Less visible or indirectly affected stakeholders include candidates with non-traditional career paths, individuals from underrepresented or historically marginalized groups, and broader labor-market participants. These stakeholders may not interact directly with the system but may

experience disproportionate impacts due to how AI-assisted screening encodes and prioritizes certain credentials, experiences, or resume formats.

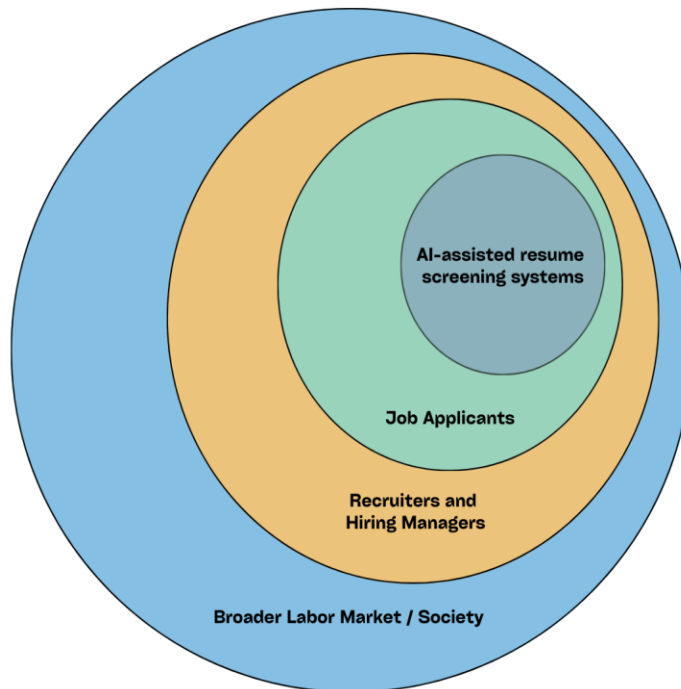


Figure 2. Stakeholder groups affected by AI-assisted resume screening systems and their relative proximity to system impacts.

4.2 Stakeholder Salience Analysis

Stakeholder salience within AI-assisted resume screening systems varies according to differences in power, legitimacy, and urgency, and these differences influence how stakeholder interests are reflected or overlooked within automated prioritization processes.

Recruiters and employing organizations typically hold the greatest power, as they control system configuration, usage practices, and final hiring decisions. Their operational needs, such as efficiency and throughput, often shape how AI tools are deployed and trusted within recruitment workflows.

Job applicants possess high legitimacy, as hiring decisions directly affect their economic opportunities and career trajectories. However, applicants have limited power and visibility within AI-assisted screening processes, particularly when system logic and ranking criteria are not transparent or contestable.

Certain stakeholder groups such as candidates with unconventional backgrounds, career gaps, or credentials that do not align neatly with standardized job descriptions, may experience reduced salience within AI-driven prioritization. Their interests may be deprioritized not by intent, but through reliance on historical data patterns or proxy indicators that favor conventional profiles.

Urgency may be unevenly distributed across stakeholders. Organizational pressures to fill roles quickly can elevate the urgency of recruiter needs while diminishing the perceived urgency of individual applicant concerns. In such contexts, AI-assisted prioritization may further reinforce existing imbalances by accelerating decision-making without sufficient opportunity for human reconsideration.

From a governance perspective, understanding these salience dynamics is critical. AI-assisted resume screening systems can inadvertently reshape stakeholder engagement by amplifying the interests of already powerful actors while reducing the visibility of stakeholders with legitimate but less quantifiable claims. Effective governance therefore requires deliberate practices to recognize, monitor, and address these imbalances to support fair and accountable hiring outcomes.

5. Risk Identification

AI-assisted resume screening systems present a range of governance risks arising from their role in shaping early-stage hiring decisions and stakeholder prioritization. These risks do not necessarily result from system malfunction, but rather from how AI outputs are generated, interpreted, and integrated into organizational decision-making processes. One key governance risk relates to fairness and bias. AI-assisted screening systems may reinforce historical hiring patterns embedded in training data or job requirements, potentially disadvantaging candidates from underrepresented groups or those with non-traditional career paths, particularly when proxy indicators such as education history, employment continuity, or keyword matching are used as substitutes for more holistic assessments of candidate suitability.

A second risk concerns transparency and explainability. Limited visibility into how resumes are evaluated or prioritized can constrain the ability of candidates and recruiters to understand, assess, or contest screening outcomes. When prioritization logic is opaque, oversight is weakened and trust in recruitment processes may be undermined. Automation bias and over-reliance represent additional governance risks. Under conditions of high application volume or time pressure, recruiters may place undue confidence in AI-generated rankings, reducing critical human review and increasing the influence of incomplete or context-insensitive system outputs on hiring decisions.

Risks related to accountability and responsibility may also arise when AI-assisted tools are embedded in recruitment workflows. Ambiguity regarding responsibility for AI-influenced

decisions can complicate efforts to address adverse outcomes, respond to stakeholder concerns, or implement corrective actions.

Finally, trust and reputational risks may emerge if AI-assisted resume screening is perceived as unfair, exclusionary, or insufficiently governed. Such perceptions can erode confidence not only in individual organizations, but also in broader hiring practices and labor-market institutions.

Collectively, these risks indicate the importance of governance practices that recognize AI-assisted resume screening as a socio-technical system, in which technical capabilities, human judgment, organizational incentives, and stakeholder expectations interact to shape outcomes.

6. Risk Impact and Likelihood Assessment

This section provides a qualitative assessment of the potential impact and likelihood of the governance risks identified in Section 5. The assessment reflects plausible risk scenarios based on typical deployment and use of AI-assisted resume screening systems in organizational hiring contexts. Risk characterizations are indicative and intended to support prioritization and governance planning rather than precise quantification.

Risks related to fairness and bias are assessed as having high potential impact and medium likelihood. When such risks materialize, they may result in discriminatory outcomes, reduced workforce diversity, and misalignment with organizational values and legal obligations. The likelihood of these risks increases when screening systems rely heavily on historical data patterns or when mitigation practices, such as bias review and human oversight, are insufficiently established.

Risks associated with transparency and explainability are assessed as having medium to high impact and medium likelihood. Limited visibility into candidate prioritization can constrain oversight, reduce stakeholder trust, and limit an organization's ability to explain or justify hiring outcomes, particularly when system documentation is limited or AI outputs are presented without contextual explanation.

Automation bias and over-reliance risks are assessed as having medium impact and medium likelihood. Under conditions of high application volume or time pressure, reliance on automated rankings may reduce critical human review and amplify the influence of system limitations on decision-making.

Risks related to accountability and responsibility are assessed as having medium impact and low to medium likelihood, depending on the clarity of organizational roles and governance structures. Where responsibility for AI-supported decisions is poorly defined, organizations may face challenges addressing adverse outcomes or responding to stakeholder concerns.

Finally, trust and reputational risks are assessed as having medium to high impact and low to medium likelihood. Although such risks may not materialize frequently, their consequences can

be significant, affecting organizational credibility, employer brand, and broader confidence in hiring practices.

Overall, this qualitative assessment underscores the importance of prioritizing governance measures that address higher-impact risks while maintaining proportional oversight across all identified risk areas.

7. Governance Controls and Mitigations

7.1 Existing Controls

AI-assisted resume screening systems are typically deployed within recruitment processes that include baseline governance controls. These controls are often embedded within broader organizational practices rather than designed specifically for AI risk management. Common existing controls include human-in-the-loop review, whereby recruiters retain responsibility for reviewing AI-generated rankings or recommendations prior to making shortlisting decisions. Recruiters are generally expected to apply professional judgment and contextual understanding rather than rely exclusively on automated outputs.

Organizations may also maintain standard recruitment policies and procedures, such as hiring guidelines, equal opportunity statements, and internal approval workflows. While not AI-specific, these policies provide a foundational governance context within which AI-assisted screening tools operate. In addition, recruitment platforms typically offer configuration options that allow organizations to adjust job criteria, skill weighting, or filtering parameters, providing a degree of operational control.

While these controls offer partial mitigation of AI-related risks, they may be insufficient if not explicitly aligned with the specific characteristics, limitations, and use conditions of AI-assisted decision-support systems. To support clearer alignment between identified risks and governance practices, Table 1 summarizes illustrative controls, mitigation measures, and indicative accountability roles.

Table 1. Governance controls and mitigation measures for identified risks.

Risk ID	Governance Objective	Control / Mitigation	Responsible Role
R-1	Reduce bias risk	Periodic bias reviews; documented evaluation criteria	HR / Governance Committee
R-2	Improve transparency	Documentation of system purpose and limitations	HR / Legal
R-3	Clarify accountability	Defined ownership of AI-supported decisions	Hiring Leadership
R-4	Mitigate automation bias	Human-in-the-loop review requirements	Recruiters

7.2 Recommended Governance Actions

To strengthen governance and mitigate identified risks, additional and more explicit controls are recommended to support responsible use of AI-assisted resume screening systems. Organizations should establish clear documentation of system purpose, scope, and limitations, including explicit statements that the system is intended to support not replace human decision-making. Such documentation should be accessible to recruiters and relevant oversight functions. Formal human oversight practices should be defined and reinforced, including expectations for recruiter review of AI outputs, guidance on appropriate use of automated rankings, and conditions under which human judgment should override system recommendations.

Organizations should implement periodic review and monitoring practices to assess potential bias, performance drift, or unintended impacts over time. These reviews may be qualitative in nature and should account for changes in job requirements, applicant populations, and labor-market conditions.

Clear roles and accountability structures should be established for AI-assisted hiring processes, including responsibility for system configuration, monitoring, escalation of concerns, and response to adverse outcomes or stakeholder inquiries.

Finally, training and awareness activities should be provided for recruiters and hiring managers to improve understanding of system capabilities, limitations, and governance expectations. Such activities support informed use and reduce the risk of automation bias. Collectively, these governance actions emphasize proportional oversight, accountability, and sustained attention to stakeholder impacts, supporting the responsible deployment of AI-assisted resume screening systems in hiring contexts.

8. Alignment with the NIST AI Risk Management Framework

This assessment is informed by the structure and principles of the NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0). The analysis aligns with the Framework's four core functions (**Govern, Map, Measure, and Manage**) to support a structured and consistent approach to identifying and addressing AI-related risks in hiring contexts.

The **Govern** function is reflected throughout the assessment through emphasis on organizational accountability, documentation, and oversight. Sections addressing purpose, scope, governance controls, and role clarity highlight the importance of establishing internal structures and practices that support responsible AI use.

The **Map** function is addressed through contextual analysis of the AI-assisted resume screening system, including system description, intended use, deployment context, and stakeholder identification. This mapping supports understanding of how the system operates within organizational and societal environments and how it may affect different stakeholders.

The **Measure** function is reflected in the qualitative identification and assessment of governance risks related to fairness, transparency, explainability, accountability, and trust. While the assessment does not quantify technical performance, it evaluates potential impacts and likelihoods to support informed risk prioritization.

The **Manage** function is addressed through the identification of existing controls and the proposal of governance actions intended to mitigate identified risks. Recommended practices emphasize human oversight, monitoring, documentation, and continuous review to support responsible system use over time.

Collectively, this alignment demonstrates how the NIST AI RMF can be operationalized in an applied governance assessment to support trustworthy deployment of AI-assisted decision-support systems in hiring contexts.

9. Monitoring and Continuous Review

Effective governance of AI-assisted resume screening systems requires ongoing monitoring and periodic review to ensure that identified risks remain understood, controlled, and proportionately addressed over time. Because organizational needs, labor-market conditions, and AI system behavior may evolve, governance practices should be adaptive rather than static.

Monitoring activities should include regular review of system use and outcomes, with attention to changes in hiring patterns, candidate demographics, and recruiter reliance on AI-generated outputs. Qualitative indicators such as recruiter feedback, candidate complaints, or observed inconsistencies in screening outcomes can provide valuable signals of emerging risks even in the absence of formal performance metrics.

Governance practices should also support periodic reassessment of system scope, assumptions, and limitations, particularly when job requirements change, recruitment volumes fluctuate, or system configurations are updated. Such reassessments help ensure that AI-assisted screening continues to align with organizational values, employment equity objectives, and stakeholder expectations.

Clear escalation and review mechanisms should be established to address identified concerns, including defined responsibility for investigating issues, documenting findings, and implementing corrective actions where appropriate. Lessons learned through monitoring and review should inform updates to documentation, training, and governance controls.

Continuous review reinforces accountability and supports responsible system use by recognizing that AI-related risks are dynamic and context-dependent. Through ongoing attention to monitoring and governance adaptation, organizations can better sustain trust in AI-assisted hiring practices over time.

10. Limitations and Reflection

This assessment is subject to several limitations that should be acknowledged. The analysis is based on publicly available information and generalized deployment practices associated with AI-assisted resume screening systems. It does not involve access to proprietary system designs, training data, internal configuration settings, or organization-specific performance metrics. As a result, the assessment does not evaluate technical model behavior or validate system outcomes.

The scope of this assessment is intentionally focused on governance, risk, and oversight considerations rather than technical or legal compliance analysis. While this approach supports structured examination of stakeholder impacts and organizational controls, deeper analysis would benefit from additional data, internal documentation, and engagement with system operators or affected stakeholders.

Future iterations of this assessment could be strengthened through more detailed examination of recruitment outcomes over time, incorporation of stakeholder feedback, and closer alignment

with organization-specific hiring practices and governance structures. Further analysis may also explore how emerging regulatory developments, evolving labor-market dynamics, and advances in AI system design influence governance requirements. Reflecting on these limitations reinforces the importance of treating AI governance as an iterative and context-dependent practice. Continuous learning, review, and adaptation are essential to maintaining responsible and trustworthy use of AI-assisted decision-support systems in hiring contexts.

11. Conclusion

AI-assisted resume screening systems play an increasingly influential role in early-stage hiring decisions, shaping how candidates are prioritized and how access to employment opportunities is mediated. As demonstrated in this assessment, the governance of such systems is not solely a technical concern, but a socio-technical responsibility that requires attention to stakeholder impacts, organizational practices, and oversight mechanisms.

Effective governance of AI-assisted hiring tools depends on clear articulation of system purpose, preservation of human judgment, and recognition of the power dynamics embedded in automated prioritization. Risks related to fairness, transparency, accountability, and trust can emerge even when systems function as intended, underscoring the need for structured and proportionate governance controls.

By applying the principles of the NIST Artificial Intelligence Risk Management Framework, this assessment illustrates how organizations can approach AI-assisted resume screening in a manner that is context-aware, stakeholder-informed, and adaptable over time. Responsible AI use in hiring contexts requires continuous monitoring, reflection, and refinement to ensure that efficiency gains do not come at the expense of equity, accountability, or public trust. Collectively, these considerations highlight the importance of embedding governance practices throughout the lifecycle of AI-assisted recruitment systems to support fair, transparent, and trustworthy hiring outcomes.